

Group 4

Lab 2: System Planning, Requirements Analysis, and  
Database Design

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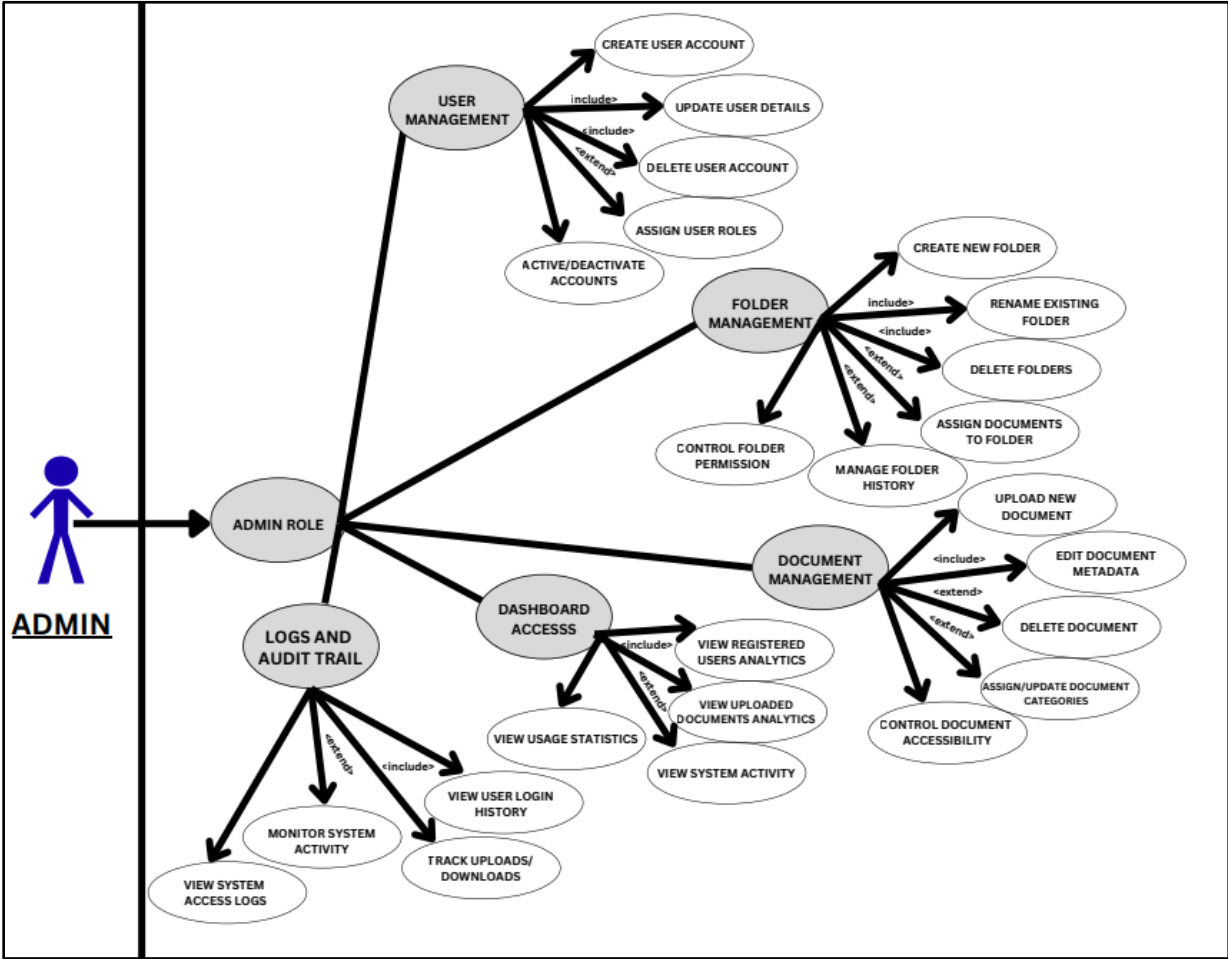
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BSCPE 2-B

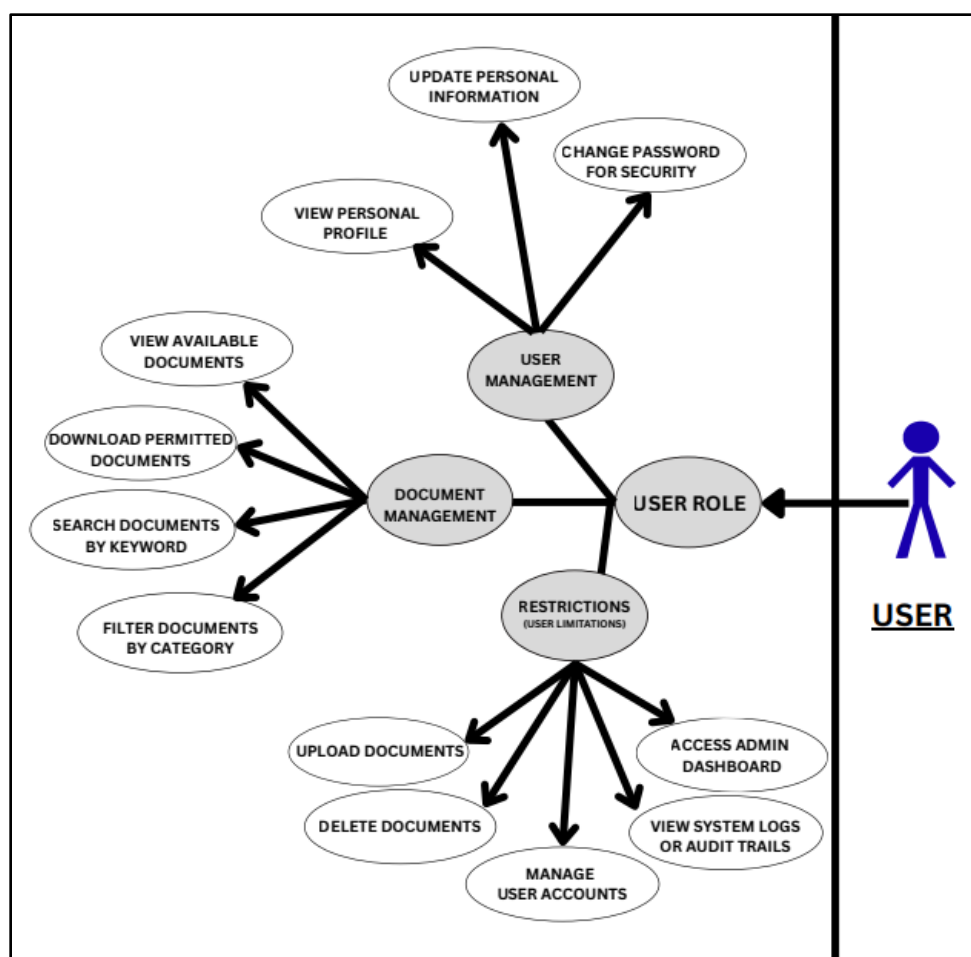
## Use case Diagram:



This diagram illustrates the Admin Role's capabilities in System:

- **User Management** – Creating, updating, and deleting user accounts, assigning roles, and activating/deactivating accounts.
- **Folder Management** – Organizing documents through folder creation, renaming, deletion, permission control, and history tracking.
- **Document Management** – Uploading documents, editing metadata, deleting files, categorizing, and controlling document accessibility.
- **Dashboard Access** – Viewing system-wide analytics such as registered users, uploaded documents, and usage statistics.
- **Logs and Audit Trail** – Monitoring login history, tracking uploads/downloads, and viewing system access logs for accountability.

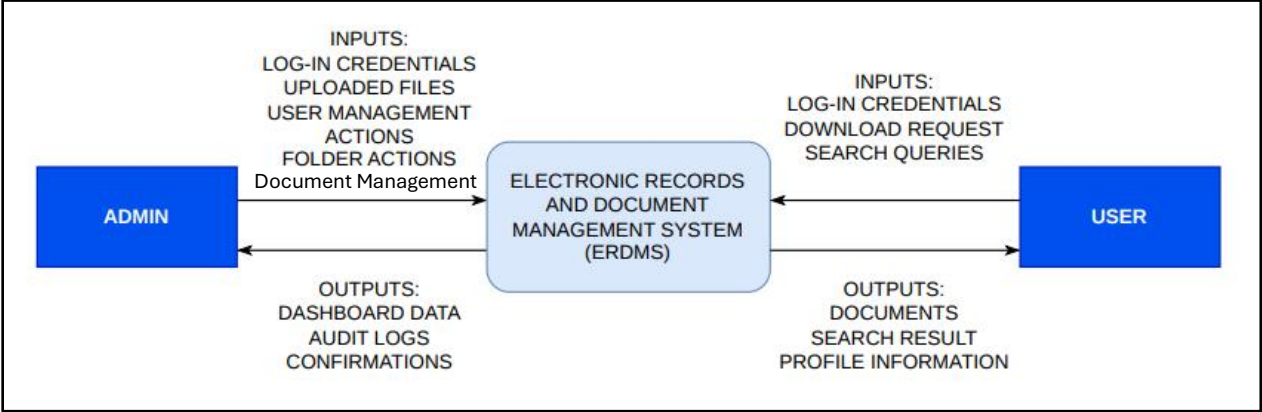
In short, the diagram shows that the Admin is the highest-level user with unrestricted access to all system functions, ensuring proper governance, security, and organization of the entire system.



This diagram illustrates the Standard User Role's capabilities and limitations in the system, showing what a regular user can and cannot do within the system.

- **User Management** – Limited to managing their own account only: viewing their personal profile, updating personal information, and changing their password.
- **Document Management** – Users can interact with documents but strictly in a read-only manner: viewing available documents, downloading permitted files, searching by keyword, and filtering by category.
- **Restrictions (User Limitations)** – The diagram explicitly shows what standard users are blocked from doing: uploading documents, deleting documents, managing user accounts, viewing system logs or audit trails, and accessing the Admin dashboard.

In short, the Standard User has a limited, controlled role focused only on accessing and retrieving documents within boundaries set by the Admin. Unlike the Admin, they cannot modify, delete, or monitor anything system-wide ensuring data security and proper access control across the system.



**Context Diagram:**

This diagram shows an overview of the system, illustrating how the two external entities (Admin and User) interact with the system as a whole, without going into internal details.

**Admin Interaction**

**Inputs (Admin → System):**

- Login credentials for authentication
- Uploaded files for document management
- User management actions (create, update, delete accounts)
- Folder actions (create, rename, delete folders)

**Outputs (System → Admin):**

- Dashboard data (analytics and system statistics)
- Audit logs (system activity tracking)
- Confirmations (feedback on actions performed)

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**User Interaction**

**Inputs (User → System):**

- Login credentials for authentication
- Download requests for permitted documents
- Search queries to find documents

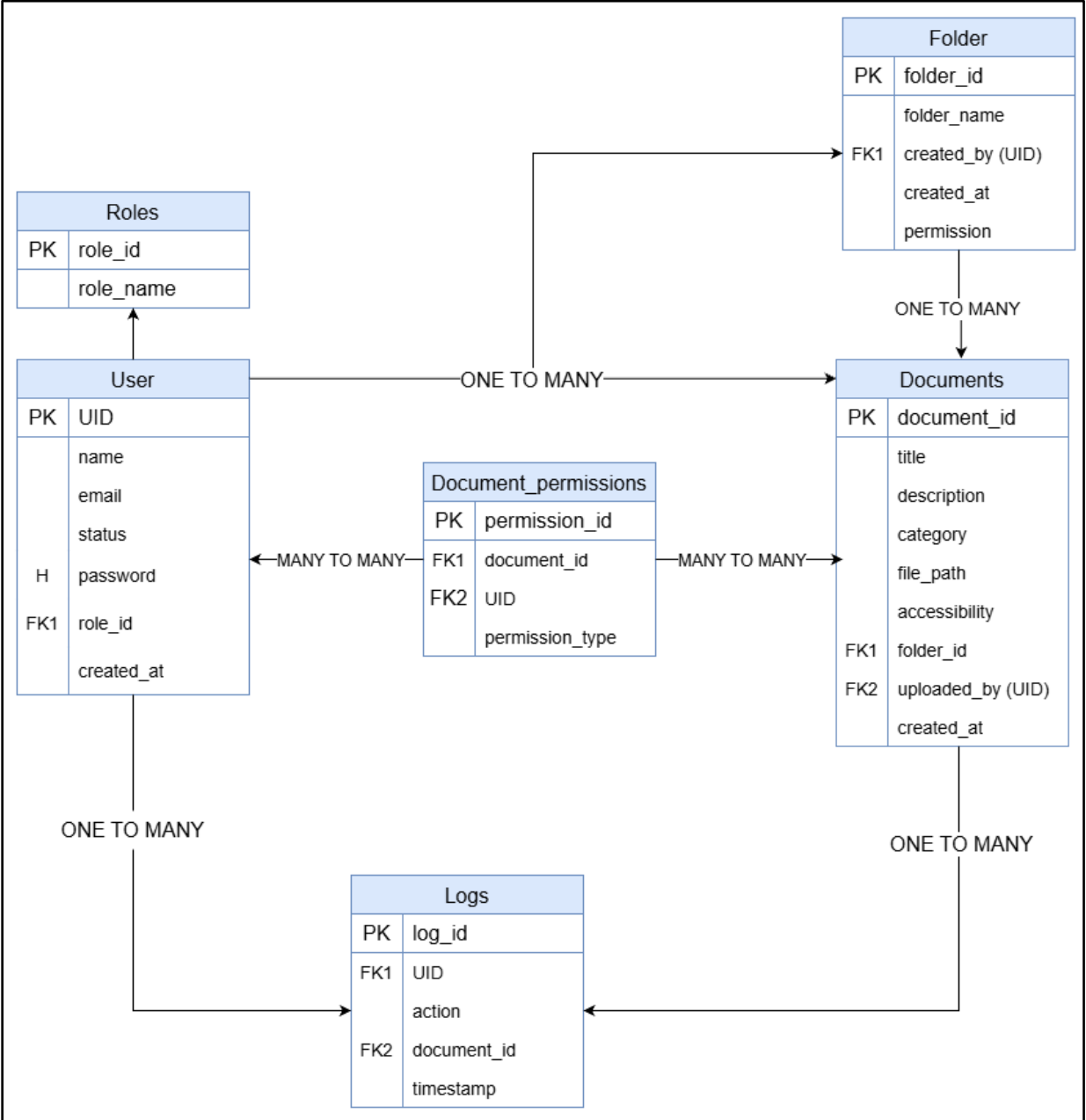
**Outputs (System → User):**

- Documents (files they are permitted to access)
- Search results (matching documents from queries)
- Profile information (their personal account details)

**Summary**

The context diagram clearly establishes the boundaries of the system, showing what goes in and what comes out for each role, without exposing internal processes. It confirms that the Admin has broader system control while the User is limited to document access and personal account interactions, which is consistent with the system requirements.

## Entity Relationship Diagram:



The ERD defines the database structure of the system, consisting of 6 tables with clearly established relationships.

**User** - serves as the central table of the system, storing essential user information including name, email, hashed password, account status, assigned role, and creation timestamp. It maintains relationships with nearly all other tables, reflecting the user's involvement across document uploads, permission assignments, and activity logging.

**Roles** - stores the role types assigned to users. Each role can belong to many users, establishing a Many-to-One relationship with the User table, which holds user credentials, status, and role assignment.

**Folder** - organizes documents within the system. A single folder can contain multiple documents, forming a One-to-Many relationship with the Documents table. The Documents table serves as the core data table, storing all file-related information including accessibility control and folder association.

**Document\_Permissions** - acts as a bridge table between Users and Documents, handling a Many-to-Many relationship that defines what specific permissions a user has over a particular document.

**Logs** - records all system activity by linking to both the User and Documents tables through One-to-Many relationships, capturing who performed what action and on which document, along with a timestamp.

Together, these tables support all core functionalities of the system.

SQL Schema:

document\_management\_sy

Tables

Views

Stored Procedures

Functions

sys

AdministrationSchemas

Information

No object selected

1CREATE DATABASE IF NOT EXISTS document\_management\_system;

2USE document\_management\_system;

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5DROP TABLE IF EXISTS Logs;

6DROP TABLE IF EXISTS Document\_permissions;

7DROP TABLE IF EXISTS Documents;

8DROP TABLE IF EXISTS Folders;

9DROP TABLE IF EXISTS Users;

10DROP TABLE IF EXISTS Roles;

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12CREATE TABLE Roles (

13role\_id INT PRIMARY KEY AUTO\_INCREMENT,

14role\_name ENUM('Admin', 'Employees') NOT NULL UNIQUE

15);

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17CREATE TABLE Users (

18user\_id INT PRIMARY KEY AUTO\_INCREMENT,

19name VARCHAR(100) NOT NULL,

20email VARCHAR(100) NOT NULL UNIQUE,

21password VARCHAR(255) NOT NULL,

22role\_id INT,

23created\_at TIMESTAMP DEFAULT CURRENT\_TIMESTAMP,

24status ENUM('active', 'inactive') DEFAULT 'active',

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26FOREIGN KEY (role\_id)

27REFERENCES Roles(role\_id)

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Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context HelpSnippets

Action Output

| # | Time     | Action                                                   | Message                                                                                                   | Duration / Fetch |
|---|----------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------|
| 1 | 14:34:25 | CREATE DATABASE IF NOT EXISTS document_management_system | 1 row(s) affected, 1 warning(s): 1007 Can't create database 'document_management_system', database exists | 0.000 sec        |
| 2 | 14:34:25 | USE document_management_system                           | 0 row(s) affected                                                                                         | 0.000 sec        |
| 3 | 14:34:25 | DROP TABLE IF EXISTS Logs                                | 0 row(s) affected                                                                                         | 0.031 sec        |
| 4 | 14:34:25 | DROP TABLE IF EXISTS Document_permissions                | 0 row(s) affected                                                                                         | 0.031 sec        |
| 5 | 14:34:25 | DROP TABLE IF EXISTS Documents                           | 0 row(s) affected                                                                                         | 0.031 sec        |
| 6 | 14:34:25 | DROP TABLE IF EXISTS Folders                             | 0 row(s) affected                                                                                         | 0.016 sec        |

Object InfoSession

document\_management\_sy

Tables

Views

Stored Procedures

Functions

sys

AdministrationSchemas

Information

No object selected

28ON DELETE SET NULL

29ON UPDATE CASCADE

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32CREATE TABLE Folders (

33folder\_id INT PRIMARY KEY AUTO\_INCREMENT,

34folder\_name VARCHAR(100) NOT NULL,

35created\_by INT,

36permissions ENUM('public', 'private', 'restricted') DEFAULT 'public',

37created\_at TIMESTAMP DEFAULT CURRENT\_TIMESTAMP,

38

39FOREIGN KEY (created\_by)

40REFERENCES Users(user\_id)

41ON DELETE SET NULL

42ON UPDATE CASCADE

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44

45CREATE TABLE Documents (

46document\_id INT PRIMARY KEY AUTO\_INCREMENT,

47title VARCHAR(255) NOT NULL,

48description TEXT,

49category VARCHAR(100),

50file\_path VARCHAR(255) NOT NULL,

51accessibility ENUM('public', 'private', 'restricted') DEFAULT 'public',

52folder\_id INT,

53uploaded\_by INT,

54created\_at TIMESTAMP DEFAULT CURRENT\_TIMESTAMP,

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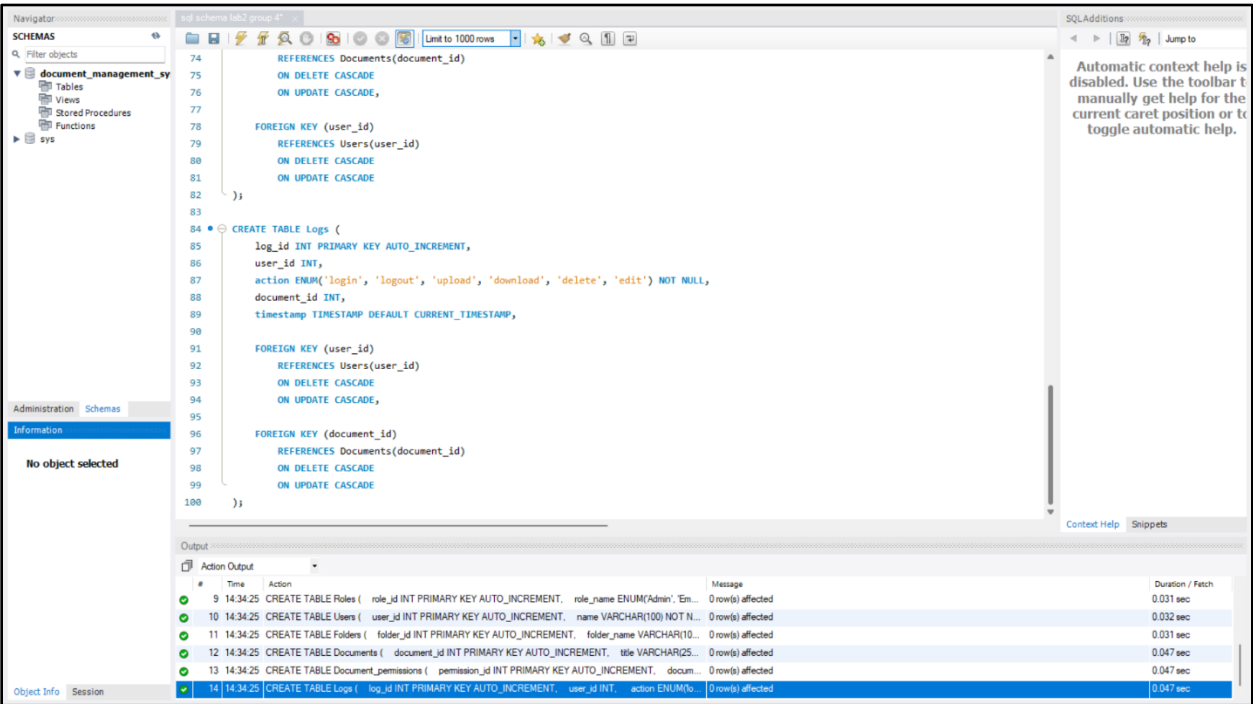
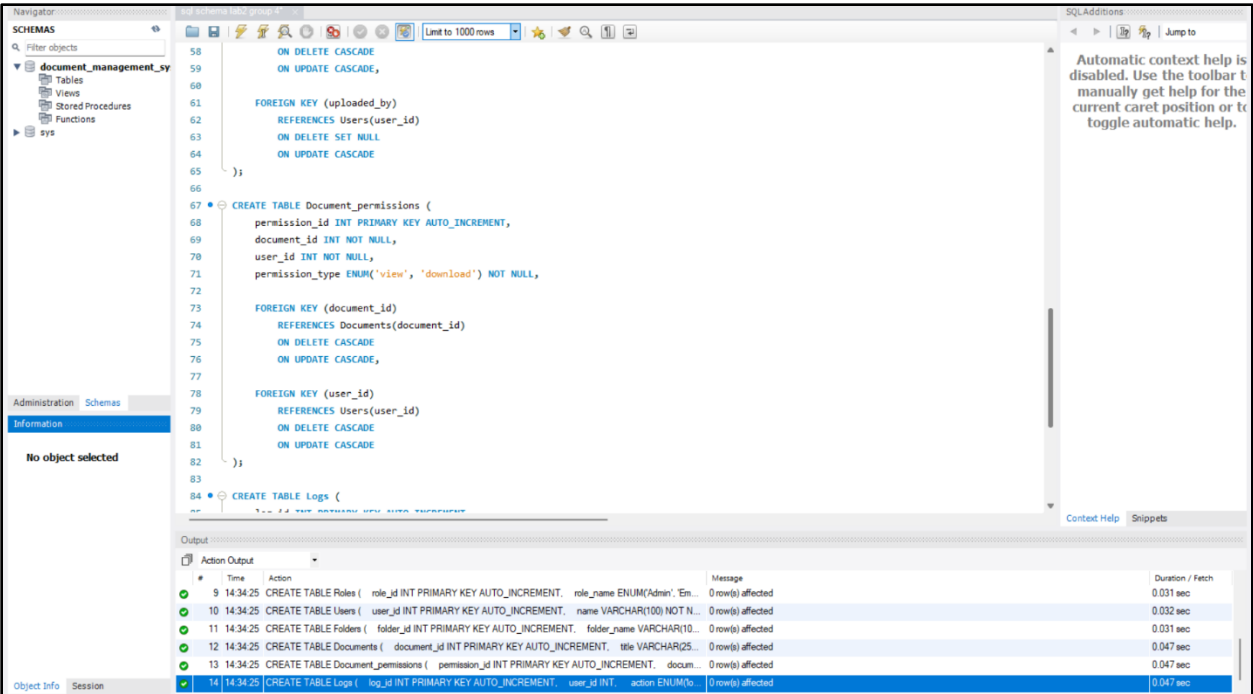
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These pictures show the SQL Schema; each picture describes the state of each field and their relationship with each other. The SQL Schema accurately portrays the ERD, since the SQL Schema is the application of the ERD to the actual database to be used.

Project Plan:

| DONE                     | PERIOD | PHASES AND TASK                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | WEEK 1 | <b>Foundations &amp; Architecture</b><br><br><b>Requirements:</b> Audit the CRD to finalize project scope and clarify technical constraints with the instructor.<br><br><b>System Design:</b> Map the data logic via an ERD and translate it into a formal SQL schema.<br><br><b>Prototyping:</b> Sketch the Angular UI layout and define the navigational flow between components.   |
| <input type="checkbox"/> | WEEK 2 | <b>Backend &amp; Database Development</b><br><br><b>Database Setup:</b> Initialize MySQL, execute schema scripts, and populate tables with dummy seed data.<br><br><b>API Construction:</b> Build a Node.js and Express server with RESTful routes for all required features.<br><br><b>Connectivity:</b> Establish a secure link between the Express backend and the MySQL database. |
| <input type="checkbox"/> | WEEK 3 | <b>Frontend &amp; System Integration</b><br><br><b>UI Build:</b> Create modular Angular components and set up services for API communication.<br><br><b>End-to-End Integration:</b> Connect the frontend to the backend to ensure real-time data flow.<br><br><b>UI Feedback:</b> Implement dynamic state management, loading indicators, and form validation.                        |
| <input type="checkbox"/> | WEEK 4 | <b>Documentation &amp; Delivery</b><br><br><b>Testing:</b> Perform rigorous functional and edge-case testing to identify and fix remaining bugs.<br><br><b>Documentation:</b> Draft technical README files, API endpoint guides, and setup instructions.<br><br><b>Presentation:</b> Finalize the local deployment environment and prepare the project demo for the instructor.       |